Financial Conditions, r^*, and the Transmission of Monetary Policy Surprises

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Abstract

High-frequency surprises identify monetary policy shocks cleanly, but their estimated effects on unemployment and inflation after 1994 are often weak or puzzling. Holding the surprise fixed, we change only its propagation. Through a nominal short rate alone, the puzzles appear. Through financial conditions as a whole, measured relative to the neutral rate r^*, a policy move that raises the two-year yield 1 percentage point raises unemployment about 0.4 point and lowers industrial production about 1 percent within a year, and lowers core prices about 0.6 percent after two years. Financial conditions matter most for unemployment; r^* for prices, whose response relies on post-2020 data.

1 Introduction

The high-frequency identification of monetary policy shocks is one of the major empirical advances in macroeconomics. Changes in interest-rate futures and Treasury yields in narrow windows around FOMC announcements isolate news about policy from other news (Kuttner, 2001; Gürkaynak et al., 2005), and their effects on asset prices are estimated precisely (Bernanke and Kuttner, 2005; Hanson and Stein, 2015; Nakamura and Steinsson, 2018; Swanson, 2021). Using these surprises to learn about the effects of policy on output, unemployment and inflation has been harder. In samples that start in the 1990s, the estimated responses are often small, imprecise or of the wrong sign (Ramey, 2016). The literature has made major progress on this problem. One line of work shows that announcements also reveal the Federal Reserve's information about the economy, so that a surprise tightening partly signals good news, and separates the two components (Romer and Romer, 2000; Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021). Another shows that surprises are partly predictable from public data that predate the announcement, and that orthogonalizing them, and adding Chair speeches, restores conventional and larger responses (Bauer and Swanson, 2023a,b). Closest to us, Gertler and Karadi (2015) show that policy surprises work through credit costs, and Caldara and Herbst (2019) that omitting credit spreads, to which policy also reacts, attenuates the response of activity. The link between futures-based surprises and monthly VARs goes back to Faust et al. (2004).

This note focuses on the other half of the exercise: how the surprise is propagated to the economy. We hold the high-frequency shock fixed and ask how much of the post-1994 puzzle reflects two features of the transmission mechanism that standard specifications leave out. First, policy affects spending through the whole configuration of financial conditions—the cost of long-term credit, the value of wealth and the exchange rate (Boivin et al., 2010)—not through a single short rate; summarizing them in a financial conditions index is common in policy analysis (Hatzius et al., 2010; Ajello et al., 2023). Second, what matters for spending and inflation is the level of real

rates relative to the neutral rate r^*, which fell substantially from the 1990s to the 2010s (Laubach and Williams, 2003; Holston et al., 2017, 2023; Hamilton et al., 2016; Kiley, 2020) and shaped the trend in yields (Bauer and Rudebusch, 2020). Over the post-1994 period, rates trended down with r^* while activity trended up; a reduced-form link between the level of rates and activity then partly reflects these opposing trends rather than the cyclical stance of policy. The two features need not matter equally for all outcomes. In standard models, activity responds to changes in financial conditions, while inflation depends on the cumulative gap between real rates and r^*, through the output gap it produces; measuring the stance relative to r^* should therefore matter most for prices.

Our design is direct. The shock is the Nakamura and Steinsson (2018) policy-news surprise, measured in narrow windows around FOMC announcements (252 announcements, 1994–2026). Its impact on the two-year yield and, when included, on a market-based financial conditions index (FCI) is read off the window, where macroeconomic variables do not move. A daily model carries the financial variables forward from the announcement day, and monthly unemployment and core inflation respond to monthly averages of the daily variables, as in a standard monthly VAR. The model respects the timing of each announcement within its month, which addresses the temporal-aggregation problem stressed by Kilian (2024) and Chudik and Georgiadis (2022); related mixed-frequency approaches include Francis et al. (2011), Ghysels (2016) and Gazzani and Vicondoa (2020). The FCI and the sequence of steps—nominal, real, and relative to r^*—are taken from a companion note on financial conditions (FCI note, 2026) and were fixed before this analysis.

The evidence builds in four steps, each with a quadratic time trend in the monthly equations. With the nominal two-year yield as the only channel, the post-1994 puzzles appear: the unemployment response is unstable across lag lengths and significantly negative with 12 lags, prices barely move, and industrial production rises. Adding the FCI turns the unemployment response conventional, large and significant, but prices fall only weakly and industrial production does not fall. Measuring rates in real terms changes little. Measuring rates and the FCI relative to r^* completes the picture: unemployment rises 0.36–0.40 percentage point at 12 months, the core price level falls 0.53–0.72 percent at 24 months, and industrial production falls about 1 percent at 12 months, all significant at the 90 percent level for every lag length. Relative to the benchmark of Bauer and Swanson (2023a), the price response is similar for two years and larger thereafter, the industrial-production response somewhat smaller, and the unemployment response about twice as large at one year and more persistent.

The robustness exercises sharpen what drives this. Financial conditions are essential for unemployment: without the FCI, no adjustment of the two-year yield gives a robust unemployment response, and with it every adjustment does. Measuring conditions relative to r^* is essential for prices: the price response is strong whenever the FCI is measured relative to the Laubach–Williams r^*, whatever the adjustment of the two-year yield, and with the nominal or real FCI only when the two-year yield is adjusted with a model-based r^*. The unemployment response is robust to the treatment of the pandemic and is larger in a sample that ends in 2019; the significant price response draws heavily on the period since 2020. The results are the same, or slightly stronger, with the orthogonalized surprises of Bauer and Swanson (2023a) or the pure-policy component of Jarociński and Karadi (2020), and with bootstrap intervals that re-estimate the daily dynamics and regenerate the monthly data recursively.

Our approach is complementary to the information-effect and predictability literatures, and it moves the burden of identification. The surprise enters our model only through the window moves of the two-year yield and the FCI and through the timing of the announcement; the rest of the propagation is the reduced-form link between financial conditions and the economy. This makes the estimates less dependent on the strength of the instrument and on its correlation with prior news, but it rests on the assumption that the economy responds to financial conditions in the same

way whatever moved them. Combining orthogonalized or decomposed surprises with propagation through financial conditions relative to r^* is a natural next step. To our knowledge, propagating high-frequency surprises through a broad financial conditions index measured relative to r^* has not been examined.

2 Data and method

Surprises and window responses. The shock is the Nakamura–Steinsson policy-news factor from the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025): the first principal component of changes in five short-term interest-rate futures around 252 FOMC announcements, February 1994 to September 2026, in the database's monetary-event window (from 10 minutes before the statement to 60 minutes after the start of the press conference, or the 30-minute statement window when there is no press conference). We scale it so that the two-year Treasury yield rises 1 percentage point in the window. The window response of the FCI is computed from the changes in the 10-year yield, the S&P 500 and the dollar index in the same windows, combined with the FCI's weights; it is 0.99 per percentage point, close to one by construction, since the weights are scaled so that a unit of the index corresponds to a policy move that raises the two-year yield 1 percentage point. Unemployment and prices are zero in the window.

Financial conditions. The FCI is the market index of FCI note (2026): a weighted sum of the 10-year Treasury yield, equity valuation (cyclically adjusted price–earnings ratio, CAPE) and the real broad dollar, with weights proportional to their responses to the same policy news. It comes in three versions, built in the same steps as in that note. *Nominal*: the variables as measured. *Real*: the 10-year yield net of 10-year expected inflation. r^**-gap* (baseline): the real 10-year yield net of the one-sided Laubach–Williams r^* (New York Fed current estimates, based on Holston et al. 2023), and equities as the gap between the earnings yield and r^*. The two-year yield follows the same rule: nominal; net of two-year expected inflation (Cleveland Fed; Haubrich et al., 2012); and net of expected inflation and r^*. Daily versions follow the daily 10-year yield, CAPE and dollar and are shifted, smoothly within each month, so that their monthly averages equal the monthly indexes exactly; the adjustments for expected inflation and r^* move slowly, so the window responses are the same in every version.

Model. The model has a daily and a monthly block. In the daily block, the two-year yield and (when included) the FCI follow a heterogeneous autoregression in their 1-, 5-, 22- and 66-day averages and respond to the current and previous month's unemployment gap and core inflation (estimated 1990–2026). In the monthly block, the unemployment gap (unemployment rate less the CBO natural rate) and core PCE inflation are rows of a monthly VAR with L lags of each and L lags of the monthly averages of the daily financial variables, estimated by OLS over February 1994–September 2026 with March–December 2020 excluded as dependent observations; L is 3, 6 or 12. Within each month macro conditions are set first, and markets respond to them; in the announcement month, macro variables have not yet moved. The response to a surprise is computed by placing the window impact on the announcement day (drawn from the empirical distribution of FOMC dates within the month), simulating the daily block forward, averaging within months, and iterating the monthly block. This is a restricted mixed-frequency VAR (Ghysels, 2016) with equal weights within the month, the restriction implied by a standard monthly VAR. A test of equal within-month weights does not reject it for unemployment; it is rejected for core inflation, although the gain in fit is very small.

Maintained trend. Every monthly equation includes a quadratic time trend. Over 1994–2026 the levels of unemployment, inflation and interest rates contain slow-moving components that

a short monthly VAR does not capture: the downward drift of rates with r^*, the low inflation of the 2010s and its rise after 2020, and shifts in the natural rate of unemployment beyond the CBO estimate. Without a trend, these components load on the lagged financial variables and mix trend with transmission. The quadratic is the most parsimonious smooth trend that allows for a U-shape in inflation. Because it is a maintained assumption, Appendix Table A3 reports all core steps without it: the nominal and real steps are much more sensitive to the trend than the r^* step, consistent with the r^* adjustment doing much of the trend's work.

Identifying assumption. The surprise affects unemployment and prices only through the variables in the daily block, and the economy responds to a given change in those variables in the same way whatever caused it. The first part is weaker the more complete the set of financial variables; the second is the assumption behind any use of financial conditions indexes to summarize the stance of policy. It is strong: equity prices, for example, partly anticipate activity. We do not decompose the FCI into components or into policy-driven and other movements: the index is designed to summarize transmission as a whole, and its weights already measure each component's response to policy news. A direct mixed-frequency regression of unemployment on the surprises that does not impose the assumption (Chudik and Georgiadis, 2022; Chudik and Kilian, 2026) gives wrong-signed point estimates, driven largely by the 2007–09 announcements, and is about ten times less precise than our estimates; it cannot discriminate between them and zero.

Inference. The baseline resamples announcements (which re-estimates the window impact and the distribution of announcement dates) and residuals of the monthly equations with a fixed design, 200 draws. Section 4 reports a fuller bootstrap that also re-estimates the daily block and regenerates the monthly data recursively. The FCI weights and r^* are held fixed throughout. We report 90 percent intervals, the convention in recent high-frequency studies (Bauer and Swanson, 2023a; Caldara and Herbst, 2019; Miranda-Agrippino and Ricco, 2021; Ramey, 2016), and also 68 percent intervals, which Jarociński and Karadi (2020) report alongside 90 percent ones.

3 Results

Table 1 reports the responses of the unemployment gap and the core PCE price level at 12, 24 and 36 months, for 3, 6 and 12 monthly lags, at each step; Figure 1 shows the four key cases.

Table 1: Responses to a policy move that raises the two-year yield 1 percentage point

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	no	0.07	−0.13	−0.27**	0.36†	0.10	−0.19	0.53**	0.29	−0.00
	yes	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
Real	no	0.14†	−0.04	−0.20**	0.33**	0.12	−0.13	0.42**	0.23	−0.01
	yes	0.44**	0.38**	0.29**	0.65**	0.65**	0.59**	0.57**	0.64**	0.64**
r^*-gap	no	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†	0.07
	yes	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
B. Core PCE price level (percent)										
Nominal	no	−0.03	−0.01	−0.02	−0.16	−0.16	−0.07	−0.26†	−0.34	−0.18
	yes	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†	−0.49†
Real	no	−0.14†	−0.08	−0.10	−0.25†	−0.23	−0.16	−0.33†	−0.37†	−0.26
	yes	−0.17†	−0.17†	−0.18†	−0.30†	−0.40†	−0.36†	−0.35†	−0.54†	−0.49†
r^*-gap	no	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
	yes	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**

Columns: horizon and number of monthly lags. **: 90 percent interval excludes zero; †: 68 percent interval excludes zero (90 percent does not). Steps as in Section 2; all specifications include a quadratic time trend. Bauer and Swanson (2023a), benchmark (read from their Figure 8): unemployment +0.19, +0.17, +0.15 at 12, 24, 36 months (90 percent band excludes zero only at 12); CPI (not core PCE) −0.36, −0.48, −0.6 (bands exclude zero).

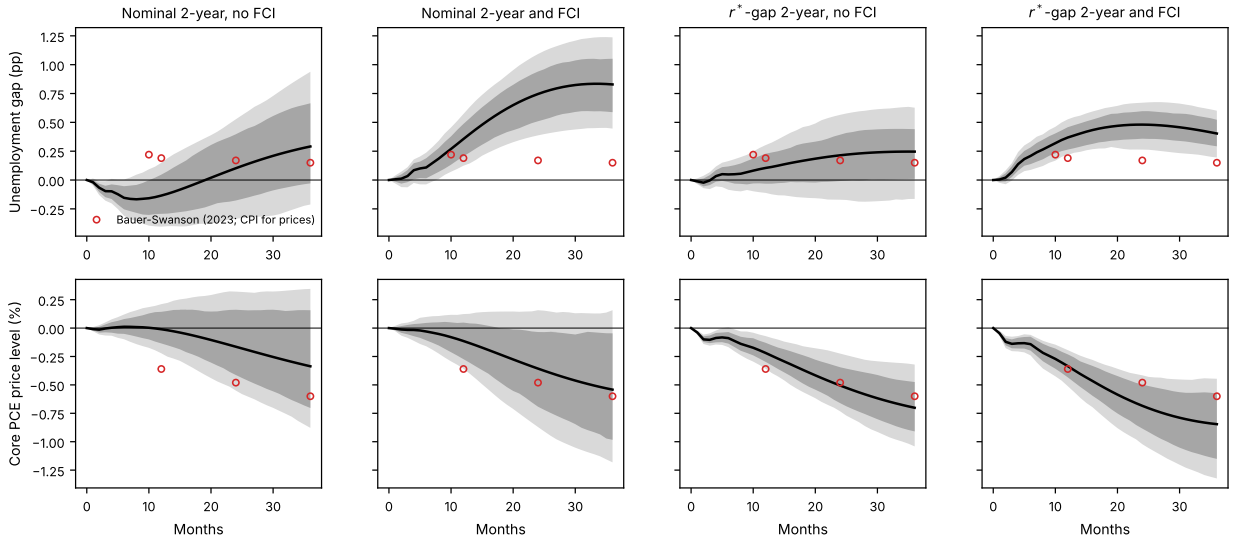


Figure 1: Responses to a policy move that raises the two-year yield 1 percentage point (6 lags). Dark band: 68 percent; light band: 90 percent. Circles: Bauer and Swanson (2023a), read from their Figure 8 (unemployment rate; CPI rather than core PCE).

Step 1: the standard set-up reproduces the puzzles. With the nominal two-year yield as the only channel—the policy indicator of Bauer and Swanson (2023a)—the unemployment response depends on the lag length and is significantly negative with 12 lags (−0.27 at 12 months), and the price level barely moves. This is the post-1994 pattern documented by Ramey (2016) and the

one Bauer and Swanson (2023a) obtain with unadjusted surprises. Industrial production rises significantly (Appendix Table A1).

Step 2: financial conditions. Adding the FCI turns the unemployment response conventional, large and significant at the 90 percent level at every horizon and lag length: 0.28–0.44 percentage point at 12 months and 0.6–0.8 at 24–36 months. The price level falls by 0.2–0.5 percent at 24–36 months, significant only at the 68 percent level and only with 6 and 12 lags, and industrial production does not fall.

Step 3: real rates. Measuring the two-year yield and the FCI net of expected inflation changes little, as in the companion note, where the nominal and real indexes are close after 1990; the price decline becomes significant at the 68 percent level at every horizon and lag length.

Step 4: rates relative to r^*. Measuring the two-year yield and the FCI relative to r^* completes the picture. Unemployment rises 0.36–0.40 percentage point at 12 months, 0.47–0.50 at 24 months and 0.35–0.43 at 36 months, nearly the same for every lag length; the core price level falls 0.33–0.34 percent at 12 months, 0.53–0.72 at 24 months and 0.57–0.97 at 36 months; all are significant at the 90 percent level for every lag length. The unemployment response is more moderate and less persistent than with the nominal FCI. Without the FCI, the r^*-gap rate gives a strong price response but a weak and lag-sensitive unemployment response (significant only with 3 lags).

Comparison with Bauer and Swanson (2023a). Their benchmark proxy VAR (1973–2020, with surprises from FOMC announcements and Chair speeches, orthogonalized to pre-announcement news) gives an unemployment response of about 0.19 percentage point at 12 months, 0.17 at 24 and 0.15 at 36 months, and a CPI decline of about 0.36, 0.48 and 0.6 percent, per percentage point on the two-year yield. Our step-4 price responses (core PCE) are similar at 12–24 months and larger at 36 months (-0.57 to -0.97); our unemployment response is about twice as large at 12 months and 2.5–3 times as large at 24–36 months, with intervals that exclude zero through 36 months where theirs include zero after about a year and a half. Their estimates use 30 years of instrument data, Chair speeches and orthogonalization; ours use FOMC windows alone and no orthogonalization.

4 Robustness

Which neutral-rate adjustment? Table 2 holds the FCI at the three indexes of the companion note—whose r^*-gap version always uses the Laubach–Williams r^*—and varies only the adjustment of the two-year yield: none (nominal), the Laubach–Williams r^* (baseline), Kiley (2020)'s r^*, a market-based measure of the nominal neutral rate—the 9–10-year forward rate from the Federal Reserve's fitted yield curve (Gürkaynak et al., 2007), which replaces both the expected-inflation and r^* adjustments—or the trailing 8-year average of the two-year yield. First, financial conditions are essential for unemployment: without the FCI, the unemployment response is weak with a model-based r^* and significantly negative with the nominal rate, the forward rate or the moving average; with any of the FCIs it is positive at every horizon and significant at the 90 percent level in almost all cases. Second, the price response is strong whenever the FCI is measured relative to r^*: with the r^*-gap FCI the price level falls 0.3–0.7 percent at 24 months whatever the adjustment of the two-year yield, including none, significant at the 90 percent level for every lag length. With the nominal or real FCI it is strong when the two-year yield is adjusted with a model-based r^* and weaker with the forward rate (whose term premium fell over the sample) or the moving average. Third, Kiley's r^* gives results close to Laubach–Williams, somewhat weaker for prices. Because the r^*-gap FCI always uses Laubach–Williams, these results show that the measurement of the

two-year yield matters less once the FCI is measured relative to r^*, not that the measure of r^* in the FCI is immaterial.

Table 2: Alternative adjustments of the two-year yield: unemployment gap (percentage points)

	12 months			24 months			36 months		
FCI	3	6	12	3	6	12	3	6	12
2-year: nominal									
none	0.07	−0.13	−0.27**	0.36†	0.10	−0.19	0.53**	0.29	−0.00
nominal	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
real	0.41**	0.32**	0.25†	0.69**	0.67**	0.57**	0.68**	0.74**	0.70**
r^*-gap	0.39**	0.31**	0.29**	0.58**	0.57**	0.55**	0.52**	0.57**	0.60**
2-year: Laubach–Williams r^*-gap (baseline)									
none	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†	0.07
nominal	0.48**	0.43**	0.35**	0.63**	0.63**	0.58**	0.49**	0.53**	0.53**
real	0.46**	0.42**	0.36**	0.59**	0.59**	0.57**	0.45**	0.49**	0.51**
r^*-gap	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
2-year: Kiley r^*-gap									
none	0.13†	−0.02	−0.15†	0.26†	0.07	−0.11	0.30**	0.13	−0.03
nominal	0.44**	0.38**	0.32**	0.58**	0.60**	0.58**	0.43**	0.50**	0.56**
real	0.44**	0.38**	0.32**	0.55**	0.58**	0.57**	0.40**	0.48**	0.54**
r^*-gap	0.39**	0.36**	0.33**	0.45**	0.48**	0.52**	0.31**	0.39**	0.45**
2-year: minus 9–10-year forward rate									
none	−0.20**	−0.20†	−0.23**	−0.25**	−0.17	−0.23†	−0.24**	−0.11	−0.14
nominal	0.26**	0.25**	0.18†	0.33**	0.42**	0.35**	0.26**	0.35**	0.35**
real	0.26**	0.26**	0.20†	0.31**	0.40**	0.35†	0.24**	0.33**	0.34†
r^*-gap	0.22**	0.23**	0.26**	0.26**	0.35**	0.40**	0.21**	0.30**	0.37**
2-year: minus trailing 8-year average									
none	−0.16†	−0.28**	−0.30**	−0.05	−0.18	−0.30†	0.02	−0.06	−0.18
nominal	0.30**	0.25**	0.21**	0.39**	0.43**	0.40**	0.27**	0.35**	0.38**
real	0.27**	0.21**	0.17†	0.36**	0.39**	0.36**	0.26**	0.33**	0.36**
r^*-gap	0.22**	0.17**	0.21**	0.33**	0.36**	0.41**	0.29**	0.37**	0.46**

See Table 1. The FCI is always one of the companion note's three indexes; the r^*-gap FCI uses the Laubach–Williams r^*. All specifications include the quadratic trend.

Table 3: Alternative adjustments of the two-year yield: core PCE price level (percent)

	12 months			24 months			36 months		
FCI	3	6	12	3	6	12	3	6	12
2-year: nominal									
none	−0.03	−0.01	−0.02	−0.16	−0.16	−0.07	−0.26†	−0.34	−0.18
nominal	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†	−0.49†
real	−0.09	−0.11	−0.13	−0.23	−0.34†	−0.33	−0.31	−0.51†	−0.50
r^*-gap	−0.20**	−0.22**	−0.21**	−0.32**	−0.50**	−0.52**	−0.32†	−0.61†	−0.72**
2-year: Laubach–Williams r^*-gap (baseline)									
none	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
nominal	−0.25**	−0.28**	−0.30**	−0.42**	−0.59**	−0.66**	−0.50**	−0.75**	−0.92**
real	−0.26**	−0.29**	−0.33**	−0.44**	−0.59**	−0.70**	−0.52**	−0.75**	−0.97**
r^*-gap	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**
2-year: Kiley r^*-gap									
none	−0.23**	−0.16†	−0.18**	−0.37**	−0.34†	−0.29†	−0.44**	−0.48†	−0.41†
nominal	−0.23**	−0.24**	−0.21†	−0.39**	−0.50**	−0.41†	−0.43**	−0.61**	−0.53†
real	−0.25**	−0.25**	−0.25**	−0.40**	−0.51**	−0.46†	−0.44**	−0.62**	−0.58†
r^*-gap	−0.34**	−0.33**	−0.29**	−0.48**	−0.62**	−0.58**	−0.47**	−0.69**	−0.68**
2-year: minus 9–10-year forward rate									
none	0.01	−0.04	−0.08	0.04	−0.04	−0.06	0.05	−0.06	−0.07
nominal	−0.05	−0.12	−0.14†	−0.07	−0.21	−0.25	−0.06	−0.22	−0.31
real	−0.06	−0.14†	−0.18†	−0.07	−0.21	−0.30†	−0.06	−0.21	−0.37
r^*-gap	−0.22**	−0.27**	−0.30**	−0.29**	−0.48**	−0.62**	−0.30†	−0.53**	−0.81**
2-year: minus trailing 8-year average									
none	−0.08	−0.06	−0.08	−0.25†	−0.27†	−0.23†	−0.37†	−0.47†	−0.45†
nominal	−0.13†	−0.15†	−0.16†	−0.27†	−0.38**	−0.39†	−0.31†	−0.49**	−0.56†
real	−0.14†	−0.15†	−0.18†	−0.28†	−0.37**	−0.42†	−0.32†	−0.49**	−0.61†
r^*-gap	−0.25**	−0.24**	−0.24**	−0.46**	−0.61**	−0.63**	−0.56**	−0.83**	−0.96**

See Table 2. The 8-year average and the Laubach–Williams trend in the two-year yield (two-year expected inflation plus r^*) are close until 2015 (averages 5.3 and 5.5 percent over 1994–2007; 2.3 and 2.5 over 2008–2015) and diverge after 2016 (1.4 and 3.6).

The pandemic and the recent period. The baseline drops March–December 2020 as dependent observations, but 2020 values still enter the 2021 equations as lags. Table 4 also drops 2021, ends the sample in December 2022, and ends it in December 2019 (estimating the daily block over the same period). With the r^*-gap rate and FCI, the unemployment response is robust: 0.26–0.31 percentage point at 12 months when 2021 is dropped, 0.32–0.34 when the sample ends in 2022, and 0.42–0.44 when it ends in 2019, significant at the 90 percent level at almost every horizon and lag length; without the FCI it is insignificant in all three samples. The price response is unchanged when the sample ends in 2022, so the data since 2023 play no special role, and it remains significant with 6 and 12 lags when 2021 is dropped (−0.43 and −0.63 percent at 24 months); but in the sample ending in 2019 it barely moves. The significant price response therefore comes from including the 2020–22 period. Figure 2 shows why the recent period is informative: after 2021 the r^*-gap two-year yield and FCI rose by about 4 points, the largest swing in the stance relative to r^* in the sample, and core inflation fell. Figure 3 drops one calendar year at a time. No single year drives the results: the unemployment response at 12 months ranges from 0.26 (dropping 2021) to 0.41, and the price level at 24 months from −0.43 (dropping 2021) to −0.86, with 90 percent intervals that exclude zero in every case. The 2020–22 dependence is thus a feature of the episode

as a whole—a large tightening relative to r^* followed by disinflation—rather than of any one year.

Table 4: Sample: r^*-gap two-year yield, without and with the r^*-gap FCI

		12 months			24 months			36 months		
Sample	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Baseline	none	0.19**	0.10	−0.03	0.34**	0.21	0.04	0.38**	0.25†	0.07
	r^*-gap	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
Also drop 2021	none	0.03	−0.03	−0.09	0.09	−0.01	−0.09	0.12	−0.01	−0.11
	r^*-gap	0.31**	0.26**	0.30**	0.38**	0.33**	0.38**	0.29**	0.28†	0.30**
End in 2022	none	0.11	0.07	−0.04	0.19†	0.11	−0.01	0.21†	0.13	0.01
	r^*-gap	0.32**	0.33**	0.34**	0.35**	0.38**	0.43**	0.23**	0.30**	0.37**
End in 2019	none	0.05	0.05	0.00	0.09	0.14	0.14	0.11	0.19	0.28
	r^*-gap	0.42**	0.42**	0.44**	0.45**	0.50**	0.60**	0.34**	0.42**	0.59**
B. Core PCE price level (percent)										
Baseline	none	−0.27**	−0.22**	−0.23**	−0.45**	−0.51**	−0.52**	−0.55**	−0.70**	−0.80**
	r^*-gap	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**
Also drop 2021	none	−0.11†	−0.09	−0.13†	−0.18†	−0.25†	−0.35**	−0.22†	−0.38†	−0.59**
	r^*-gap	−0.16†	−0.21**	−0.28**	−0.24†	−0.43**	−0.63**	−0.29†	−0.60**	−0.93**
End in 2022	none	−0.25**	−0.20**	−0.21**	−0.42**	−0.49**	−0.51**	−0.51**	−0.70**	−0.83**
	r^*-gap	−0.34**	−0.33**	−0.30**	−0.51**	−0.68**	−0.71**	−0.56**	−0.87**	−1.02**
End in 2019	none	0.02	0.02	0.08†	0.06	0.02	0.03	0.10	0.02	−0.06
	r^*-gap	−0.02	−0.03	−0.05	0.02	−0.00	−0.12	0.03	−0.03	−0.22†

See Table 1. "End in" samples also estimate the daily block over the shorter period.

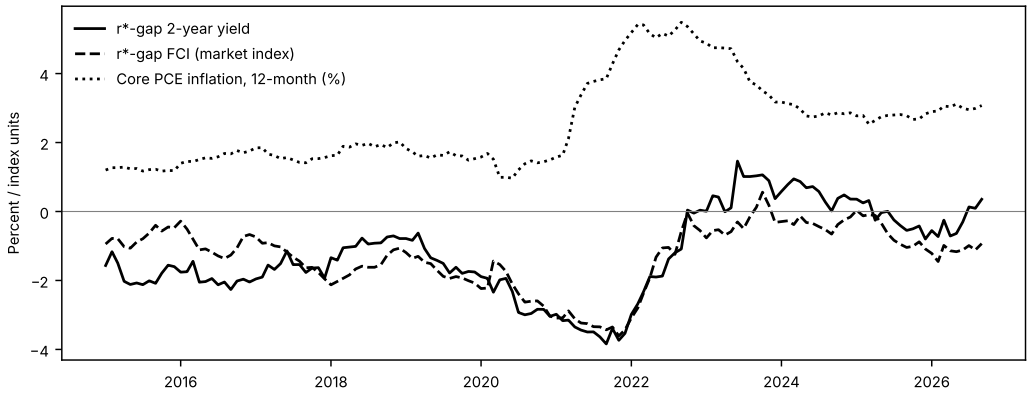


Figure 2: The r^*-gap two-year yield, the r^*-gap FCI and 12-month core PCE inflation, 2015–2026.

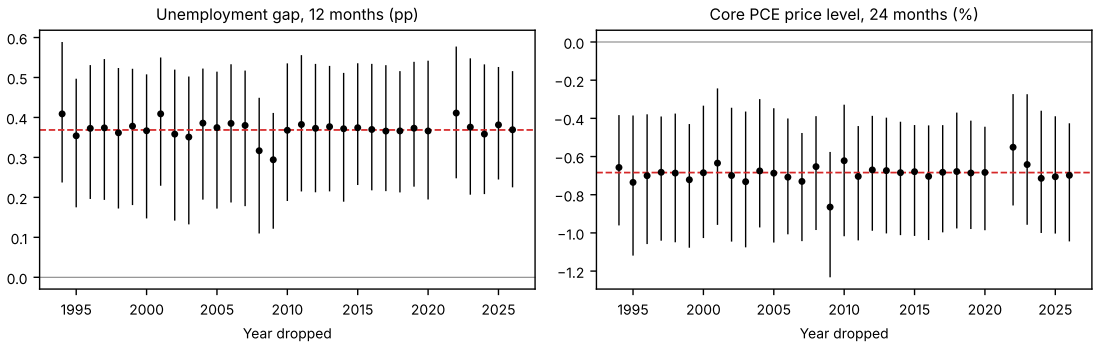


Figure 3: Leave-one-year-out: baseline (r^*-gap two-year yield and FCI, 6 lags, 100 draws), dropping each calendar year in turn from the monthly equations. Bars: 90 percent intervals; dashed line: full-sample estimate.

Industrial production. Appendix Table A1 adds industrial production to the monthly block. Without the FCI, industrial production rises after a tightening with the nominal and real rates and is insignificant with the r^*-gap rate. With the nominal or real FCI it does not fall significantly. Only at the r^* step does it fall, by 0.84–1.05 percent at 12 months, significant at the 90 percent level for every lag length, against about 1.5 percent in Bauer and Swanson (2023a); it recovers within about three years. In exploratory work, adding a commodity price index as well produced rising commodity prices after a tightening, a reminder that the propagation assumption is least plausible for the most forward-looking, demand-sensitive variables.

Alternative shocks. Appendix Table A2 replaces the Nakamura–Steinsson surprise with three alternatives, keeping everything else unchanged: the surprise purged of Greenbook forecasts, the orthogonalized surprise of Bauer and Swanson (2023a) (their posted series for FOMC announcements, 201 of our events through 2019), and the "poor man's" sign restriction of Jarociński and Karadi (2020), which keeps only announcements where the two-year yield and the S&P 500 move in opposite directions (157 events). In our design a shock enters only through the window moves, scaled to a 1 point rise in the two-year yield, and the timing of announcements, so this exercise checks the ratio of the FCI's window move to the two-year yield's: 0.99 for the baseline, 1.07 for the Bauer–Swanson surprise and 1.09 for the Jarociński–Karadi selection. The last ratio is higher partly by construction, since that selection keeps announcements where equities, which enter the FCI with a negative sign, fall when yields rise. With these ratios, the responses are slightly larger than the baseline: at the r^* step, unemployment rises 0.40–0.45 percentage point at 12 months and the price level falls 0.55–0.77 percent at 24 months. The Greenbook-purged surprise gives results within 0.02 of the baseline. Without the FCI, the three shocks give essentially identical responses. The results are therefore not sensitive to how the surprise is cleaned of information or predictable components, which matter here only through the window ratio.

Inference. Appendix Table A4 reports a fuller bootstrap with 1,000 draws that, in addition to resampling announcements, re-estimates the daily block on a moving-block resample of its regressions (66-day blocks) and regenerates the monthly data recursively from the estimated equations with Rademacher-signed residuals before re-estimating them. The intervals are essentially unchanged: at the r^* step every significance marker is the same, and elsewhere a few markers change between the 68 and 90 percent levels. The FCI weights and r^* remain fixed.

Other checks. In exploratory versions of the model, adding a Baa–Treasury spread to the daily block lowered the unemployment response by a third to a half; because no intraday credit series exists, the spread's window response could not be measured cleanly, and we do not pursue it here, but

credit is the most important omitted channel (Gilchrist and Zakrajšek, 2012; Caldara and Herbst, 2019). Replacing the model's daily dynamics with the directly estimated post-announcement path of the financial variables for the first one to two months gives larger but much less precise responses.

5 Conclusion

High-frequency surprises identify monetary policy shocks well; the difficulty has been in tracing their effects on the economy. Holding the surprise fixed and changing only its propagation, we find that two standard features of the transmission mechanism account for much of the post-1994 puzzle. Policy works through financial conditions as a whole, and it is the level of rates and financial conditions relative to a slow-moving neutral rate that matters for spending and inflation. Propagating window-identified surprises through a broad financial conditions index, measured relative to r^*, gives conventional, precise and lag-robust responses of unemployment, prices and industrial production, of a size comparable to the best recent estimates.

The two ingredients play different roles. Financial conditions are essential for the unemployment response, in every variant and sample we consider. Measuring financial conditions relative to r^* is essential for the price response: with the FCI measured relative to the Laubach–Williams r^*, the price response is strong whatever the adjustment of the two-year yield. The unemployment response is robust to the pandemic; the price response draws heavily on the period since 2020, when the stance relative to r^* moved most. These findings complement the information-effect and predictability literatures; the mechanisms can operate together, and combining orthogonalized or decomposed surprises with our propagation is a natural next step. The main limitations are the identifying assumption—that policy reaches the economy through the financial variables in the model, and that the economy responds to them the same way whatever moved them—the maintained quadratic trend, the omission of credit spreads, and intervals that do not include uncertainty about r^* or the FCI weights. The broader lesson is that the decline in r^* over the past three decades is not only a fact about the level of rates but also a measurement problem for estimating the effects of policy.

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A Additional tables

Table A1: Monthly block with industrial production

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	no	0.07	−0.04	−0.19†	0.26†	0.16	−0.04	0.26	0.30†	0.19
	yes	0.41**	0.36**	0.22**	0.60**	0.59**	0.47**	0.51**	0.56**	0.49**
Real	no	0.12	0.03	−0.14†	0.24†	0.15	−0.02	0.24†	0.23	0.15
	yes	0.41**	0.37**	0.24**	0.54**	0.53**	0.46**	0.43**	0.48**	0.45**
r^*-gap	no	0.16†	0.16†	0.03	0.30**	0.27†	0.14	0.33**	0.30†	0.25†
	yes	0.44**	0.42**	0.34**	0.52**	0.50**	0.50**	0.42**	0.42**	0.42**
B. Industrial production (percent)										
Nominal	no	1.06**	1.26**	1.65**	1.03†	1.26†	1.64**	1.16†	1.00†	1.10†
	yes	−0.16	−0.29	0.11	−0.09	−0.04	0.15	0.45	0.39	0.43
Real	no	0.64†	0.77**	1.13**	0.72	0.97†	1.28**	0.87†	0.85†	0.91†
	yes	−0.33	−0.52	−0.23	−0.11	−0.11	−0.01	0.42	0.31	0.34
r^*-gap	no	0.13	−0.04	0.27	0.13	0.23	0.32	0.31	0.44	0.21
	yes	−0.84**	−1.05**	−0.95**	−0.46†	−0.48	−0.80**	0.18	0.08	−0.21

See Table 1. Price-level responses are similar to Table 1. Bauer and Swanson (2023a), industrial production: about −1.5, −1.0 and −0.85 at 12, 24 and 36 months, read from their Figure 8.

Table A2: Alternative shocks, with the FCI

		12 months			24 months			36 months		
Shock	Step	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nakamura–Steinsson	nominal	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83**	0.77**
	r^*-gap	0.40**	0.37**	0.36**	0.47**	0.48**	0.50**	0.35**	0.40**	0.43**
Greenbook-purged NS	nominal	0.43**	0.34**	0.27**	0.75**	0.73**	0.62**	0.75**	0.81**	0.76**
	r^*-gap	0.39**	0.36**	0.34**	0.46**	0.47**	0.48**	0.34**	0.39**	0.42**
Bauer–Swanson orth.	nominal	0.47**	0.39**	0.32**	0.81**	0.79**	0.67**	0.81**	0.87**	0.80**
	r^*-gap	0.44**	0.41**	0.40**	0.52**	0.52**	0.54**	0.40**	0.44**	0.46**
Jarociński–Karadi	nominal	0.48**	0.40**	0.33**	0.82**	0.81**	0.69**	0.83**	0.89**	0.81**
	r^*-gap	0.45**	0.42**	0.41**	0.53**	0.54**	0.55**	0.41**	0.45**	0.47**
B. Core PCE price level (percent)										
Nakamura–Steinsson	nominal	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.54†	−0.49†
	r^*-gap	−0.34**	−0.33**	−0.33**	−0.53**	−0.68**	−0.72**	−0.57**	−0.85**	−0.97**
Greenbook-purged NS	nominal	−0.08	−0.11	−0.11	−0.24	−0.35†	−0.30	−0.33	−0.54†	−0.48
	r^*-gap	−0.34**	−0.33**	−0.33**	−0.52**	−0.67**	−0.71**	−0.56**	−0.83**	−0.95**
Bauer–Swanson orth.	nominal	−0.09	−0.12	−0.12	−0.25	−0.37†	−0.32	−0.34	−0.55†	−0.51
	r^*-gap	−0.36**	−0.35**	−0.35**	−0.55**	−0.71**	−0.76**	−0.59**	−0.88**	−1.01**
Jarociński–Karadi	nominal	−0.09	−0.12	−0.12	−0.25	−0.37†	−0.33	−0.34	−0.56†	−0.51
	r^*-gap	−0.36**	−0.36**	−0.35**	−0.55**	−0.72**	−0.77**	−0.60**	−0.89**	−1.03**

See Table 1. All shocks scaled to raise the two-year yield 1 percentage point in the window. Step: the nominal or r^*-gap two-year yield and FCI.

Table A3: Core steps without the quadratic trend

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	no	−0.13†	−0.27**	−0.34**	−0.01	−0.19†	−0.36**	0.08	−0.08	−0.25†
	yes	0.18**	0.09	0.07	0.38**	0.29**	0.19†	0.42**	0.37**	0.24**
Real	no	−0.02	−0.14†	−0.25**	0.07	−0.10	−0.28**	0.12	−0.03	−0.21†
	yes	0.25**	0.18†	0.12	0.40**	0.33**	0.24**	0.38**	0.36**	0.26**
r^*-gap	no	0.08	0.03	−0.08	0.16†	0.04	−0.15†	0.20†	0.05	−0.16
	yes	0.30**	0.32**	0.34**	0.38**	0.39**	0.41**	0.34**	0.38**	0.37**
B. Core PCE price level (percent)										
Nominal	no	0.01	0.06	0.01	−0.08	0.03	0.05	−0.17	−0.07	0.03
	yes	−0.09	−0.07	−0.10	−0.28†	−0.24†	−0.23	−0.42†	−0.41†	−0.36†
Real	no	−0.10†	−0.02	−0.07	−0.19†	−0.08	−0.06	−0.26†	−0.16	−0.08
	yes	−0.17†	−0.14†	−0.18**	−0.33†	−0.32†	−0.31†	−0.44†	−0.47**	−0.41†
r^*-gap	no	−0.15†	−0.13†	−0.19**	−0.22†	−0.26†	−0.31†	−0.28†	−0.39	−0.44†
	yes	−0.18†	−0.25**	−0.33**	−0.28†	−0.48**	−0.61**	−0.37†	−0.68**	−0.84**

As Table 1, without the trend.

Table A4: Core steps with the fuller bootstrap

		12 months			24 months			36 months		
Step	FCI	3	6	12	3	6	12	3	6	12
A. Unemployment gap (percentage points)										
Nominal	no	0.07	$-0.13^{\dagger}$	-0.27^{**}	$0.36^{\dagger}$	0.10	$-0.19^{\dagger}$	0.53^{**}	0.29	-0.00
	yes	0.44^{**}	0.36^{**}	0.28^{**}	0.76^{**}	0.75^{**}	0.63^{**}	0.76^{**}	0.83^{**}	0.77^{**}
Real	no	$0.14^{\dagger}$	-0.04	-0.20^{**}	0.33^{**}	0.12	$-0.13^{\dagger}$	0.42^{**}	0.23	-0.01
	yes	0.44^{**}	0.38^{**}	0.29^{**}	0.65^{**}	0.65^{**}	0.59^{**}	0.57^{**}	0.64^{**}	0.64^{**}
r^*-gap	no	0.19^{**}	0.10	-0.03	0.34^{**}	$0.21^{\dagger}$	0.04	0.38^{**}	$0.25^{\dagger}$	0.07
	yes	0.40^{**}	0.37^{**}	0.36^{**}	0.47^{**}	0.48^{**}	0.50^{**}	0.35^{**}	0.40^{**}	0.43^{**}
B. Core PCE price level (percent)										
Nominal	no	-0.03	-0.01	-0.02	-0.16	-0.16	-0.07	-0.26	-0.34	-0.18
	yes	-0.09	-0.11	-0.11	-0.24	$-0.36^{\dagger}$	$-0.31^{\dagger}$	-0.33	$-0.54^{\dagger}$	$-0.49^{\dagger}$
Real	no	$-0.14^{\dagger}$	-0.08	-0.10	$-0.25^{\dagger}$	$-0.23^{\dagger}$	-0.16	$-0.33^{\dagger}$	$-0.37^{\dagger}$	-0.26
	yes	$-0.17^{\dagger}$	$-0.17^{\dagger}$	$-0.18^{\dagger}$	$-0.30^{\dagger}$	$-0.40^{\dagger}$	$-0.36^{\dagger}$	$-0.35^{\dagger}$	$-0.54^{\dagger}$	$-0.49^{\dagger}$
r^*-gap	no	-0.27^{**}	-0.22^{**}	-0.23^{**}	-0.45^{**}	-0.51^{**}	-0.52^{**}	-0.55^{**}	-0.70^{**}	-0.80^{**}
	yes	-0.34^{**}	-0.33^{**}	-0.33^{**}	-0.53^{**}	-0.68^{**}	-0.72^{**}	-0.57^{**}	-0.85^{**}	-0.97^{**}

As Table 1, 1,000 draws: announcements resampled; daily block re-estimated on a moving-block resample (66-day blocks); monthly data regenerated recursively from the estimated equations with Rademacher-signed residuals and re-estimated. r^* and the FCI weights held fixed.